

RVCC TECHNICAL ARCHITECTURE & AUDIT

Technology Stack, Section Handling & Engineering Recommendations

A comprehensive breakdown of all apps, architectural choices (pros & risks), and the roadmap for scalability and performance.

ARCHITECTURE	APPS COUNT	THEME / PALETTE	STATUS
Turborepo + Edge API	4 Discrete Services	White & Marine Blue	Production Ready / Scaling

1 Technology Stack & Handling By Application

 apps/web

Marketing + Enquire (:3000)

- **Next.js 16 (Turbopack) & React 19:** Server-rendered marketing pages and optimized dynamic routing.
- **TailwindCSS v4 & Lenis:** Lightning-fast styling engine with inertial smooth scrolling.
- **Three.js & @react-three/fiber:** Interactive WebGL 3D model visualization.
- **Framer Motion & PageFlip:** Micro-animations and flip-book brochure catalog experience.
- **PDF.js & React-PDF:** Client-side compliant document previewer.
- **Zod & reCAPTCHA:** Multi-step registration validation & anti-bot shielding.

Next.js 16 Three.js Tailwind v4 Zod Vercel

 apps/api

Unified Backend API (:4000)

- **Hono v4 on Cloudflare Workers:** Ultra-low latency edge compute runtime (sub-millisecond cold starts).
- **Postgres.js:** Direct high-performance serverless PostgreSQL connection driver.
- **Cloudflare R2 + aws4fetch:** S3-compatible zero-egress asset storage for compliance documents.
- **Worker-Mailer & Nodemailer:** Edge email dispatch for registration verifications and OTPs.
- **Dual-Target Execution:** Cloudflare Workers for production + Node.js HTTP server for dev.

Hono v4 Cloudflare Workers R2 PostgreSQL

 apps/admin

Staff Backoffice (:3001)

- **Next.js 16 & React 19:** Authenticated backoffice with session token proxies (`src/proxy.ts`).
- **Recharts:** KPI metric cards, registration trends, and real-time activity analytics.
- **React-PDF:** Integrated compliance certificate verification and vendor audit panels.
- **Lucide Icons & Flag Icons:** Visual status badges, geographic distribution indicators.
- **Custom Migration Scripts:** TSX-driven administration and account seeding CLI.

Recharts Session Auth Tailwind v4 Vitest

 apps/vendor

Supplier Self-Service (:3002)

- **Next.js 16 Protected Portal:** Isolated supplier workspace for profile management.
- **Zod Form Validation:** Capability declarations, machinery specs, and certificate submissions.
- **Signed R2 Uploads:** Secure direct-to-bucket upload flows for heavy supplier documentation.
- **Lenis & Framer Motion:** Clean, polished UI with step-based status trackers.

Supplier Portal R2 Uploads Framer Motion Zod

Shared Infrastructure & Network Topology **Central PostgreSQL Database & Cloudflare R2 Storage**

Single source of truth for vendor registries, enquiry leads, audit trails & media buckets

▲ REST API, JWT Sessions & Direct R2 Presigned URLs via Edge API (Hono) ▲

apps/web

Customer Enquiries (:3000)

apps/vendor

Vendor Self-Service (:3002)

apps/admin

Staff Backoffice (:3001)

2 What We Did Right vs. What Needs Attention

✓ WHAT WE DID RIGHT (STRENGTHS)

1. Discrete Turborepo App Isolation

Splitting **Marketing**, **Vendor**, and **Admin** into separate Next.js apps prevents administrative code and sensitive business logic from leaking into public bundles, while allowing autonomous team deployments.

Impact: Enhanced security and optimized frontend bundle sizes.

2. Edge Compute with Hono & Workers

Deploying the unified API (`apps/api`) on Cloudflare Workers using Hono delivers sub-millisecond cold starts and distributed edge execution with zero server maintenance overhead.

Impact: Global performance with near-zero infrastructure cost.

3. Cloudflare R2 Storage without Egress Fees

Serving heavy supplier documents, certificates, and media through Cloudflare R2 eliminates AWS S3 egress penalties while maintaining full S3 API compatibility.

Impact: 100% predictable, low-cost asset hosting.

4. Modern UI Stack (Tailwind v4 + Framer)

Using Tailwind CSS v4 alongside Framer Motion and Lenis gives the platform modern micro-interactions, smooth scrolling, and a premium aesthetic without heavy CSS frameworks.

Impact: State-of-the-art user experience across devices.



WHAT IS WRONG / RISKY (VULNERABILITIES)

1. Direct Edge DB Connections (Connection Storms)

Cloudflare Workers connecting directly to standard PostgreSQL via `postgres.js` will exhaust `max_connections` under burst traffic, triggering intermittent **503 Service Unavailable** errors.

Fix: Connect via Cloudflare Hyperdrive or PgBouncer.

2. No Shared Packages (`packages/` folder)

All 4 apps duplicate TypeScript types, DTOs, and Zod schemas independently. When an enquiry or vendor data model changes, developers must manually synchronize code across 3 repositories.

Fix: Extract `packages/types` and `packages/schemas`.

3. Raw SQL Queries (Lack of ORM Type Safety)

Writing hand-crafted SQL strings inside API route handlers introduces maintenance friction, schema drift, and increases the potential risk of SQL syntax bugs during migrations.

Fix: Adopt Drizzle ORM for zero-overhead type-safe queries.

4. Heavy SSR Overkill in Admin & Vendor Apps

Next.js App Router with server-side rendering is ideal for the public marketing site (`web`), but adds proxy and hydration complexity for internal, authenticated backoffices.

Fix: Leverage React Query (TanStack Query) for client caching.

3 Strategic Improvements & Priority Roadmap

PRIORITY	INITIATIVE	ACTION PLAN	EXPECTED RESULT
P0 - URGENT	Connection Pooler / Hyperdrive	Route Cloudflare Workers database traffic through Cloudflare Hyperdrive or Supabase/Neon transaction pooling.	Eliminates DB connection exhaustion & crashes at peak load.
P0 - URGENT	Extract Shared Monorepo Packages	Create packages/schemas and packages/types to share Zod validation rules and DTOs across Web, Admin, and API.	Zero duplicate validation logic; 100% type sync between frontend & backend.
P1 - HIGH	Drizzle ORM Migration	Replace raw SQL strings in apps/api/src/modules/ with Drizzle schema declarations and queries.	Automatic migrations, type-safe database queries, and easier refactoring.
P1 - HIGH	TanStack Query (React Query)	Implement React Query in apps/admin and apps/vendor for data fetching, table pagination, and optimistic updates.	Instant dashboard navigation, background data sync, and cleaner UI code.
P2 - MEDIUM	Centralized Telemetry & Sentry	Integrate edge error monitoring (Sentry / Axiom) across Cloudflare Workers and Next.js frontends.	Real-time alerts on failed enquiry submissions or API exceptions.